^{2}H(^{34}S,pγ) 1973Wa10

^{34}S(d,p)^{35}S in inverse kinematics.

1973Wa10: A 59.6-MeV ^{34}S beam was produced from the BNL MP-tandem Van de Graaff facility. Targets were 200-μg/cm^2 TiD prepared by evaporating titanium onto the Cu, Al, and Mg target backings in a deuterium atmosphere. γ rays were detected using a 35-cm^3 Ge(Li) detector at 0° with FWHM=2 keV at 656 keV. Measured Eγ. Deduced $T_{1/2}$ for two ^{35}S levels using Doppler Shift Attenuation lineshape analysis.

^{35}S Levels

E(level)†	J$^\pi$‡	$T_{1/2}$#	Comments
0	3/2$^+$		
1574	1/2$^+$	2.29 ps *35*	$T_{1/2}$: Lifetime=3.3 ps *5* from 1973Wa10.
2350	3/2$^-$	0.90 ps *14*	$T_{1/2}$: Lifetime=1.3 ps *2* from 1973Wa10.

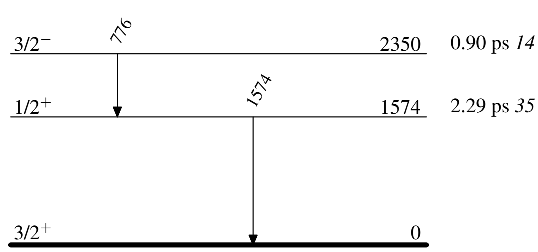
† From 1973Wa10.
‡ From the Adopted Levels.
# From Doppler Shift Attenuation Method (DSAM).

γ(^{35}S)

E_γ	E_i(level)	J_i^π	E_f	J_f^π
776	2350	3/2$^-$	1574	1/2$^+$
1574	1574	1/2$^+$	0	3/2$^+$

^{2}H(^{34}S,pγ) 1973Wa10

Level Scheme



$^{35}_{16}S_{19}$